

Arrays in VB.Net

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Level: Two

One-Dimensional Arrays

Ex: Declare a 5-element array with of integers:

```
Dim A(5) As Integer
```

```
A(1) = 75
```

```
A(2) = 80
```

```
A(3) = 70
```

```
A(4) = 85
```

```
A(5) = 90
```

Or

```
Dim A() As Integer = {75, 80, 70, 85, 90}
```

One-Dimensional Arrays

Example 1: Write a program in VB. Net to read array B(5) then find the largest element in this array and multiply the elements by any Number using For ... Next statement?

Example 1

The image shows a screenshot of a Windows application window titled "Form1". The window has a standard Windows XP-style title bar with minimize, maximize, and close buttons. The main content area is divided into three vertical sections. On the left is a large white rectangular area labeled "ListBox1" in blue text. On the right is another large white rectangular area labeled "ListBox2" in blue text. In the center, between the two list boxes, is a vertical stack of six buttons: "Enter the Numbers", "Find the Max", "Muliply by" (note the typo), "clear", "Next", and "Exit".

Example 1 cont.

```
1 reference
Public Class Form1
    Dim B(5) As Integer
    Dim i As Integer

    0 references
    Private Sub Button1_Click(sender As Object, e As EventArgs) Handles Button1.Click
        For i = 0 To 4
            B(i) = InputBox("Enter the Array Elements: ")
            ListBox1.Items.Add(B(i))
        Next
    End Sub

    0 references
    Private Sub Button2_Click(sender As Object, e As EventArgs) Handles Button2.Click
        Dim max As Integer = B(0)
        For i = 1 To 4
            If B(i) > max Then
                max = B(i)
            End If
        Next
        MessageBox.Show("The Maximum Number is: " & max)
    End Sub

    0 references
    Private Sub Button3_Click(sender As Object, e As EventArgs) Handles Button3.Click
        ListBox1.Items.Clear()
        ListBox2.Items.Clear()
    End Sub
```

Example 1 cont.

```
0 references
Private Sub Button4_Click(sender As Object, e As EventArgs) Handles Button4.Click
    Form2.Show()
    Me.Hide()
End Sub

0 references
Private Sub Button5_Click(sender As Object, e As EventArgs) Handles Button5.Click
    End
End Sub

0 references
Private Sub Button6_Click(sender As Object, e As EventArgs) Handles Button6.Click
    Dim M As Integer
    Dim w(5) As Integer
    M = InputBox("Enter the M Number: ")
    For i = 0 To 4
        w(i) = B(i) * M
        ListBox2.Items.Add(w(i))
    Next
End Sub
End Class
```

Two-Dimensional Arrays

Example 1: Design the following form and write a program in VB. Net to read two arrays $A(2,2)$, $B(2,2)$ then find the Multiplication of elements between arrays A and B in new array $C(2,2)$ using For .. Next Statement?

Example 1

The image shows a Windows application window titled "Form2". Inside the window, there are three empty rectangular boxes arranged horizontally, each labeled above with "Array A", "Array B", and "Array C" in red text. Below these boxes, there are four buttons arranged horizontally: "Sum", "clear", "Previous", and "Exit". The window has a standard Windows title bar with minimize, maximize, and close buttons.

Example 1 cont.

```
1 reference
Public Class Form2
    0 references
    Private Sub Button1_Click(sender As Object, e As EventArgs) Handles Button1.Click
        Dim I, J As Integer
        Dim A(2, 2), B(2, 2), C(2, 2) As Integer
        For I = 1 To 2
            For J = 1 To 2
                A(I, J) = InputBox("Enter The Values of Array A ")
                RichTextBox1.Text &= (A(I, J) & " ")
            Next
            RichTextBox1.Text &= (Environment.NewLine)
        Next
        For I = 1 To 2
            For J = 1 To 2
                B(I, J) = InputBox("Enter The Values of Array B ")
                RichTextBox2.Text &= (B(I, J) & " ")
            Next
            RichTextBox2.Text &= (Environment.NewLine)
        Next
        For I = 1 To 2
            For J = 1 To 2
                C(I, J) = A(I, J) * B(I, J)
                RichTextBox3.Text &= (C(I, J) & " ")
            Next
            RichTextBox3.Text &= (Environment.NewLine)
        Next
    End Sub
```

Example 1 cont.

0 references

```
Private Sub Button2_Click(sender As Object, e As EventArgs) Handles Button2.Click
    RichTextBox1.Clear()
    RichTextBox2.Clear()
    RichTextBox3.Clear()
End Sub
```

0 references

```
Private Sub Button3_Click(sender As Object, e As EventArgs) Handles Button3.Click
    Form1.Show()
    Me.Hide()
End Sub
```

0 references

```
Private Sub Button4_Click(sender As Object, e As EventArgs) Handles Button4.Click
    End
End Sub
End Class
```

Thank You